

Air University



Programming Fundamentals (Lab)

Year 2024

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Question 01Code

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      // Declaring variables
6      int num, dif, no_terms, term;
7
8      // Taking values from user
9      cout << "Enter a specific starting number. " << endl;
10     cin >> num ;
11     cout << "Enter the Common difference. " << endl;
12     cin >> dif;
13     cout << "Enter the number of terms. " << endl;
14     cin >> no_terms;
15     cout << "\nThe Arithmetic series is " << endl;
16
17     // Using for loop for printing the results
18     for (int i = 0; i < no_terms; i++)
19     {
20         term = num + i * dif;
21         cout << endl << term;
22     }
23     return 0;
24 }
```

Output

```
D:\University\Programming Fundamentals (Lab)\Assignments\Assignment 2\Question 1.exe
Enter a specific starting number.
3
Enter the Common difference.
4
Enter the number of terms.
7

The Arithmetic series is

3
7
11
15
19
23
27
-----
Process exited after 20.65 seconds with return value 0
Press any key to continue . . .
```

Question 02

Code

```
1  #include <iostream>
2  #include <cmath>
3  using namespace std;
4  int main()
5  {
6      // Declaring Variables
7      int num, ratio, no_terms, series;
8
9      // Taking values from user
10     cout << "Enter starting number for Geometric series. " << endl;
11     cin >> num;
12     cout << "Enter the common ratio. " << endl;
13     cin >> ratio;
14     cout << "Enter No. of Terms. " << endl;
15     cin >> no_terms;
16
17     cout << "\nThe Geometric Series is " << endl;
18
19     // Using for loop for series
20     for ( int i = 0 ; i < no_terms ; i++ )
21     {
22         series = num * pow ( ratio, i ) ;
23         cout << endl<< series ;
24     }
25     return 0;
26 }
```

Output

```
D:\University\Programming Fundamentals (Lab)\Assignments\Assignment 2\Question 2.exe
Enter starting number for Geometric series.
3
Enter the common ratio.
5
Enter No. of Terms.
6

The Geometric Series is

3
15
75
375
1875
9375
-----
Process exited after 8.833 seconds with return value 0
Press any key to continue . . .
```

Question 03

Code

```
1 #include <iostream>
2 using namespace std;
3 int main ()
4 {
5     // Declaring variables
6     int num1=0, num2=1, no_terms, next_term, counter=0;
7     cout << "Enter No. Terms for Fibonacci Series. " << endl;
8     cin >> no_terms ;
9     cout << "\nFibonacci series are. " << endl;
10
11     // Using while loop for the series
12     while ( counter < no_terms )
13     {
14         cout << num1 << " ";
15         next_term = num1 + num2 ;
16         num1 = num2 ;
17         num2 = next_term ;
18         counter++ ;
19     }
20
21     return 0;
22 }
```

Output

```
D:\University\Programming Fundamentals (Lab)\Assignments\Assignment 2\Question 3.exe
Enter No. Terms for Fibonacci Series.
15

Fibonacci series are.
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
-----
Process exited after 4.876 seconds with return value 0
Press any key to continue . . .
```

Thank You for Your Time!

The End